

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

AI

Practical – III

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Class: TYIT	Batch: 01
Date of Assignment: 18/7/26	Date/Time of Submission: 18/7/26

Q.1 In a college:

20% of students know Python.

80% of students know Java.

90% of Python students get placed.

40% of Java students get placed.

A randomly selected student is placed.

Find:

Probability that the student knows Python.

Probability that the student knows Java.

Probability that a placed student knows Python.

CODE:

Probabilities

P_Python = 0.20

P_Java = 0.80

P_Placed_Python = 0.90

P_Placed_Java = 0.40

Total probability of being placed

P_Placed = (P_Placed_Python * P_Python) + (P_Placed_Java * P_Java)

Bayes Rule

P_Python_Placed = (P_Placed_Python * P_Python) / P_Placed

P_Java_Placed = (P_Placed_Java * P_Java) / P_Placed

print("Probability student knows Python:", P_Python)

print("Probability student knows Java:", P_Java)

print("Probability placed student knows Python:", round(P_Python_Placed, 4))

print("Probability placed student knows Java:", round(P_Java_Placed, 4))

print("Soham Acharekar T001")

OUTPUT:

```
Probability student knows Python: 0.2
Probability student knows Java: 0.8
Probability placed student knows Python: 0.36
Probability placed student knows Java: 0.64
Soham Acharekar T001
```

Q.2 In a city:

85% of taxis are Green.

15% are Blue.

A witness correctly identifies the taxi color 80% of the time.

A witness claims the taxi involved in an accident was Blue. What is the probability that the taxi was actually Blue?

CODE:

```
# Probabilities
P_Blue = 0.15
P_Green = 0.85
P_Correct = 0.80
P_Wrong = 0.20
# Probability witness says Blue
P_Witness_Blue = (P_Correct * P_Blue) + (P_Wrong * P_Green)
# Bayes Rule
P_Blue_WitnessBlue = (P_Correct * P_Blue) / P_Witness_Blue
print("Probability taxi was actually Blue:", round(P_Blue_WitnessBlue, 4))
print("Soham Acharekar T001")
```

OUTPUT:

```
Probability taxi was actually Blue: 0.4138
Soham Acharekar T001
```